



UTM
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Assignment 1

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Introduction

Employee attrition is the gradual reduction in a company's workforce due to employees leaving and not being replaced immediately or at all. It's a natural process that occurs in every organization, but it can become problematic if the attrition rate is too high.

Understanding the factors that lead to employee turnover is essential for developing effective retention strategies and improving workplace satisfaction.

This study focuses on analysing employee-related data to discover patterns that contribute to attrition and performance variations. Our group aims to identify the key reasons why employees leave a company and what factors influence their job performance. We will analyse various attributes in the dataset such as age, job satisfaction, monthly income, performance rating, work life balance, years at the company and other factors also.

The results of the analysis will help the business better understand employee behavior, increase retention of employees, and improve worker productivity in overall. Human resource management may improve overall business performance, save hiring and training expenses, and increase employee retention by using data-driven choices. By conduct this analysis, businesses can also proactively engage employees by improving areas such as work-life balance, compensation strategies, and career development paths.

Dataset Overview

The IBM HR Analytics Employee Attrition & Performance dataset comprises 1470 rows and 35 columns. Each of the data represent various attributes related to employee personal information, job-related details, experience and tenure, compensation, and satisfaction and performance metrics. The focus of this dataset is to helps HR departments predict attrition and identify key factors influencing it.

Column Description

Target Variable

- **Attrition:** Indicates whether the employee has left the company (Yes) or is still employed (No).

Employee Personal Information

- **Age:** Employee's age in years.
- **Over18:** Indicates if the employee is over 18 years old. (Always 'Y')
- **Gender:** Male or Female.
- **MaritalStatus:** Single, Married, or Divorced.
- **Education:** Education level (1='Below College', 2='College', 3='Bachelor', 4='Master', 5='Doctor').
- **EducationField:** Field of study (e.g., Life Sciences, Medical, Marketing, Technical Degree, Human Resources, Other).
- **DistanceFromHome:** Distance from the employee's home to the workplace

Employee Job-Related Information

- **Department:** Department within the company (e.g., Sales, Research & Development, Human Resources).
- **JobRole:** Specific role of the employee (e.g., Sales Executive, Research Scientist).
- **JobLevel:** Job level within the organization. (1='Entry level', 2='Junior level', 3='Mid-level', 4='Senior level', 5='Executive').
- **EmployeeNumber:** Unique ID assigned to each employee.
- **EmployeeCount:** Always 1 (not useful – constant).
- **StandardHours:** Always 80 (not useful – all full-time).
- **OverTime:** Whether the employee works overtime (Yes/No).
- **BusinessTravel:** Frequency of business travel (e.g Non-Travel, Travel_Rarely, Travel_Frequently).

Employee Experience and Tenure

- **TotalWorkingYears:** Total years of professional experience.
- **YearsAtCompany:** Number of years employee at the current company.
- **YearsInCurrentRole:** Years employee has spent in the current role.
- **YearsWithCurrManager:** Years employee has worked with the current manager.
- **YearsSinceLastPromotion:** Years since employee's the last promotion.
- **NumCompaniesWorked:** Number of companies the employee has worked for.

Employee Compensation Details

- **MonthlyIncome:** Monthly income of the employee.
- **MonthlyRate:** Monthly wage rate.
- **HourlyRate:** Hourly wage.
- **DailyRate:** Daily wage.
- **PercentSalaryHike:** Percentage increase in salary.
- **StockOptionLevel:** Stock option level granted (0 = None, 3 = High).

Employee Performance and Satisfaction and Engagement

- **PerformanceRating:** Performance rating (1='Low', 2='Good', 3='Excellent', 4='Outstanding').
- **JobSatisfaction:** Job satisfaction level (1='Low', 2='Medium', 3='High', 4='Very High')
- **JobInvolvement:** Level of involvement in the job (1='Low', 2='Medium', 3='High', 4='Very High').
- **EnvironmentSatisfaction:** Satisfaction with the work environment (1='Low', 2='Medium', 3='High', 4='Very High').
- **RelationshipSatisfaction:** Satisfaction with relationships at work (1='Low', 2='Medium', 3='High', 4='Very High').
- **WorkLifeBalance:** Work-life balance rating (1='Bad', 2='Good', 3='Better', 4='Best').
- **TrainingTimesLastYear:** Number of training sessions attended in the last year.

Business insight

1. Overtime Strongly Influences Attrition

Employees who frequently work overtime are more likely to leave the company. One possible reason is that working long hours regularly can lead to stress and tiredness. This could indicate burnout or dissatisfaction due to work-life imbalance. To reduce this issue, the company should manage workloads better, for example hire more staff, or allow flexible working hours to support employee well-being.

2. Low Job Satisfaction Increases Attrition Risk

Employees with low job satisfaction scores tend to leave more often. When employees are not happy with their work, they may feel unmotivated and eventually decide to leave the company. This insight emphasizes the importance of regular feedback mechanisms, manager-employee communication, and job enrichment strategies to keep employees motivated and satisfied.

3. High Attrition Among Younger Employees

Younger employees (lower age range) tend to switch careers frequently. This group might seek faster career growth, more challenging roles, or competitive offers elsewhere. To retain this demographic, companies should offer dynamic career paths, learning opportunities and regular feedback systems tailored to younger professionals.

4. Impact of Training on Employee Retention

Employees who are continually provided with training appear slightly more committed to the company. Continuous learning opportunities can make individuals feel developed and committed. Companies should consider investing in employee development programs and skills seminars to improve employees and promote retention.

5. Monthly Income is Positively Correlated with Total Working Years

Analysis of the dataset reveals a strong positive correlation between an employee's total working years and their monthly income. This states that employees with more year experience tend to earn higher salaries because they have more skills with standard industry practices. In a business perspective, this analysis indicates that employees that have more years experience have higher labour costs, which should be balanced with their contribution and performance to optimize workforce efficiency and budget planning.

6. Low-Level Employees Are More Likely to Quit

Attrition is more likely to occur among workers with lower job levels, shorter tenures, lower monthly wages, and fewer years of overall job satisfaction. It also means that workers in basic level may feel undervalued, feeling bored, or have limited opportunities for promotion, which might lead to their resignation. Organizations should invest in the onboarding process, guidance, and intern development programs for staying junior talent and reduce early attrition rates.

Conclusion

This analysis has provided us the valuable insights into employee attrition dynamics and the various factors influencing job performance and satisfaction. By analyzing a full HR dataset, we could identify significant factors such as overtime work, job satisfaction, age, training, compensation and job level that have a crucial bearing on employee retention. The findings highlight the necessity to promote a good work-life balance, clear career development prospects and a culture of continuous learning and feedback.

Empowered with these insights, organizations can proactively address areas of possible risk and increase employee engagement. Focused HR initiatives such as expediting employee growth opportunities, building internal mobility and creating personalized retention plans can be launched to reduce attrition rates and build a more stable and productive workforce. In today's competitive job market, businesses that make data-driven decisions to prioritize employee satisfaction and well-being will be more formidable competitors in the war to retain talent, reduce turnover costs and drive long-term organizational success.